Production Technology

Module 1

Kinematics of Machine Tools

- **Kinematics of machine tools** deals with the study of motion (without considering forces) in machines used for manufacturing operations. It focuses on how cutting tools and workpieces move relative to each other to produce the desired shape, size, and surface finish.

1. Basic Concepts

- ◆ **Relative Motion**
- Material removal occurs due to **relative motion** between:
- Cutting tool
- Workpiece
- Example:
- In a **lathe**, the workpiece rotates and the tool feeds linearly.
- In a **milling machine**, the cutter rotates and the workpiece feeds.

2. Types of Motions in Machine Tools

- **Primary (Cutting) Motion**
  - Responsible for cutting action.
  - Usually rotary or linear.
  - Provides cutting speed.
- **Feed Motion**
  - Moves tool/work gradually for continuous cutting.
  - Slower than primary motion.
- **Auxiliary Motions**
  - Tool positioning
  - Clamping
  - Return stroke (in shaping)

3. Kinematic Elements of Machine Tools

- **Kinematic Chain**
  - Sequence of links and joints transmitting motion from motor to tool.
  - Example: Motor → Gearbox → Spindle → Tool
- **Kinematic Pair**
  Two links connected with relative motion:
  - Sliding pair (lathe carriage)
  - Turning pair (spindle bearings)
- **Transmission Systems**
  - Gear trains
  - Belt drives
  - Chain drives
  - Lead screws

4. Speed and Feed Relationships

- **Cutting Speed (V)**

- $V = \pi DN/1000$ (m/min)

  D = Diameter (mm)

  N = RPM

- **Feed Rate**

- Lathe: mm/rev
- Milling: mm/tooth
- Drilling: mm/rev

5. Kinematic Structures in Common Machine Tools

- **Lathe**
  - Workpiece rotation → Cutting motion
  - Tool linear feed → Feed motion
- **Milling Machine**
  - Cutter rotation → Cutting motion
  - Table movement → Feed motion
- **Drilling Machine**
  - Drill rotation → Cutting motion
  - Axial feed → Feed motion
- **Shaper**
  - Reciprocating tool → Cutting motion
  - Workpiece feed → Feed motion

6. Stepped vs. Stepless Speed Regulation

- **Stepped (Geared Head)**
  - Discrete speed levels
  - Gearbox-based
- **Stepless (Variable Speed Drive)**
  - Continuous speed variation
  - Hydraulic or electronic control (e.g., CNC)

7. Importance of Kinematics in Machine Design

- Determines surface finish
- Influences tool life
- Controls dimensional accuracy
- Affects productivity
- Ensures proper power transmission

Selection of Speed Range

- **Step 1: Determine Cutting Speed Limits**

- Cutting speed depends on:
- Work material (steel, aluminum, cast iron, etc.)
- Tool material (HSS, Carbide, Ceramic, etc.)
- Type of operation (turning, milling, drilling)
- Typical range example:
- Mild steel (HSS tool): 20–40 m/min
- Aluminum (Carbide tool): 150–300 m/min

- **Step 2: Convert Cutting Speed to Spindle Speed**
- $N=1000V\pi/DN$ Where:
- N = Spindle speed (rpm)
- V = Cutting speed (m/min)
- D = Diameter (mm)
- From minimum and maximum cutting speeds → calculate:
- Nmin and Nmax

Step 3: Determine Speed Range Ratio

- R= Nmax/ Nmin
- This is called **range ratio**.
- Typical machine tools:
- Lathe: 6 – 12
- Milling: 8 – 18
- Drilling: 4 – 8

Geometric Progression of Speeds

- Machine tool speeds are arranged in **geometric progression (GP)**.
- N1,N2,N3,…,Nz
- $N_k = N_1 \phi^{(k-1)}$

Where:

- ϕ= Step ratio
- z = Number of speed steps

- **Step Ratio Formula**
- $\Phi = R^{1/(z-1)}$ Standard step ratios:
- 1.06
- 1.12
- 1.26
- 1.41
- 1.58

- **Selection of Feed Range**
- Feed is selected based on:
- Surface finish requirement
- Tool strength
- Machine rigidity
- Power available
- Feed is also arranged in geometric progression.

Machine	Feed Range
Lathe	0.05 – 2 mm/rev
Milling	0.02 – 1 mm/tooth
Drilling	0.05 – 0.5 mm/rev

Design Considerations

- **Factors Affecting Speed Range**

- Work diameter variation
- Tool material
- Production rate
- Machine power

- **Factors Affecting Feed Range**

- Required surface roughness
- Tool wear
- Material hardness
- Depth of cut

- **Importance of Proper Selection**
- Prevents tool breakage
- Ensures good surface finish
- Improves production rate
- Avoids overloading motor
- Reduces vibration

Layout of Speeds in Machine Tools

- The **layout of speeds** refers to the systematic arrangement of spindle speeds in a machine tool gearbox so that the required range of speeds is obtained using a minimum number of gears and shafts.
- It is an important part of **machine tool kinematic design**.

- **Purpose of Speed Layout**
- To obtain required **minimum and maximum spindle speeds**
- To distribute speeds in **geometric progression**
- To design an economical gearbox
- To ensure smooth transmission of power

- **Steps in Speed Layout Design**

Step 1: Determine Speed Range

- $R=N_{max}/N_{min}$

Step 2: Decide Number of Speed Steps (z)

Based on:

- Type of machine (lathe, milling, drilling)
- Required flexibility
- Production type

Step 3: Calculate Step Ratio (ϕ)

- $\phi = R^{1/(z-1)}$
- Speeds are arranged as:
- $N_1,\ N_{1\phi},\ N_{1\phi}{}^{2},\ \ldots,\ N_1\phi^{z-1}$

Types of Speed Layout

1. Arithmetic Layout (Not preferred)

- Speeds differ by constant value
- Not economical
- Rarely used

2. Geometric Layout (Preferred)

- Speeds differ by constant ratio
- Most efficient
- Used in all standard machine tools

- **Structural Formula of Speed Layout**
- Number of speeds:
- $z=p_1 \times p_2 \times p_3 \ldots z$
- Where:
- p_1, p_2, p_3 = Number of speed changes in each stage
- Example:

If

z=12 Possible structural formula:

- $12=3 \times 2 \times 2$ This means:
- Stage 1 → 3 speed steps
- Stage 2 → 2 speed steps
- Stage 3 → 2 speed steps

- **Ray Diagram (Speed Layout Diagram)**
- A **ray diagram** shows:
- Arrangement of speed stages
- Distribution of transmission ratios
- Gearbox structure
- Features:
- Vertical lines → Shafts
- Slanted lines → Gear ratios
- Intersections → Speed combinations
- It helps in:
- Selecting gear teeth numbers
- Designing gearbox arrangement

- **Example**
- Given:
- N_{min}=125 rpm N_{max}=2000 rpm
- z=5
- Range ratio:
- R=2000/125=16
- Step ratio:
- $\phi=16^{1/4}=2$
- Speeds: 125,250,500,1000,2000
- Structural layout:
- 5=5×1

Design Considerations

- Minimum number of gears
- Uniform load distribution
- Compact gearbox
- Ease of shifting
- Minimum power loss

Importance

- Ensures proper cutting conditions
- Improves machine efficiency
- Reduces manufacturing cost
- Increases tool life

- **Graphical Representation of Speed and Structural Diagrams (Machine Tool Gearbox Design)**
- In machine tool kinematics, **graphical methods** are used to design and visualize spindle speed arrangements. The two important diagrams are:
- **Speed Diagram (Ray Diagram)**
- **Structural Diagram**
- These are essential in gearbox design for lathes, milling, drilling machines, etc.

- **Speed Diagram (Ray Diagram)**
- 🔹 **Purpose**
- To represent the **distribution of spindle speeds**
- To show how speeds are obtained through different gearbox stages
- To visualize geometric progression of speeds

- **Basis**
- Speeds follow **geometric progression**:
- $N_k = N_1 \phi^{(k-1)}$

Where:

- ϕ = step ratio
- z = number of speeds

- **Construction Procedure**
- Draw vertical lines → represent **shafts**
- Mark speed levels vertically (in GP order)
- Connect speeds using slant lines → represent **gear ratios**
- Each stage has its own set of rays

Example

If:

- $z=12=3\times2\times2$

Then:

- Stage 1 → 3 steps
- Stage 2 → 2 steps
- Stage 3 → 2 steps

- In the ray diagram:
- First shaft → motor input
- Intermediate shafts → speed changing stages
- Final shaft → spindle
- Each slanted line represents a gear pair.

- **Characteristics**
- Speeds arranged logarithmically
  Lines diverge progressively
  Helps in selecting transmission ratios
  Ensures uniform speed distribution

- **Structural Diagram**
- 🔹 **Purpose**
- Shows:
- Arrangement of speed-changing groups
- Number of gears in each stage
- Order of transmission
- It is a simplified representation derived from the ray diagram.

◆ Structural Formula

$$z = p_1 \times p_2 \times p_3$$

Example:

$$12 = 3 \times 2 \times 2$$

Structural notation:

$$12(1:3)(3:6)(6:12)$$

Where:

- Each bracket represents one stage
- Numbers show speed multiplication

- **Structural Diagram Representation**
- Horizontal lines → shafts
- Vertical grouping → stages
- Circles → gears
- Lines → power transmission
- It gives:
- Number of shafts required
- Gear arrangement
- Speed group sequence

Speed Diagram	Structural Diagram
Shows actual speed distribution	Shows gearbox structure
Based on geometric progression	Based on speed grouping
Used to select transmission ratio	Used to design gearbox layout
Graphical speed levels	Block representation